

Other Blood and Immune Disorders in Chronic Kidney Disease

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IMMUNE DYSFUNCTION

Patients with chronic kidney disease (CKD) have a high morbidity and mortality from infection. This is partially due to immune dysfunction, although multiple other factors are involved.

Bacterial Infections

Infections are a major cause of hospitalization in patients with CKD; infection-related hospitalization rates are at least three to four times (nondialysis CKD) or eight times (dialysis) higher than in individuals with normal kidney function.¹ The number of days spent in hospital for infections has slightly decreased over the last decade. Nevertheless, they are still higher than those for cardiovascular (CV) disease (4 vs. 2.2 days per patient year).¹ Infection is also an important cause of mortality, accounting for 8% of deaths in dialysis patients. The risks of pulmonary infections and infections of the genitourinary system both increase with decreasing kidney function. For example, patients with stage 4 CKD have a 15-fold higher risk for severe pulmonary infections compared with those with an estimated glomerular filtration rate (eGFR) greater than 60 mL/min/1.73 m². Furthermore, mortality from pneumonia is markedly enhanced by concurrent CKD,² which was reflected in the extremely high mortality among patients undergoing hemodialysis (HD) during the COVID-19 pandemic³ (see [Chapter 59](#)).

In HD patients, bloodstream infections are another major cause of infection-related hospitalization, particularly in patients with central venous catheters.⁴ Catheter-based HD is among the major risk factors for the development of bacterial endocarditis. More than 50% of cases of bacteremia are caused by *Staphylococcus aureus*. Nevertheless, there is also a relevant rate of gram-negative bacteremia in dialysis patients, indicating that contamination or infection of the dialysis access is not the only cause of bloodstream infection.

The high incidence of bacterial infection in CKD patients may be one clinical consequence of immune dysfunction; another is atypical clinical presentations of infections, such as the lack of fever in 20% to 40% of kidney patients with bacteremia. When bacteremia is suspected, blood cultures should be obtained frequently. Another aspect of immune dysfunction is the high rate of false-negative tuberculin (Mantoux) skin tests with anergic skin reactions in the presence of positive interferon- γ release assays, indicating impaired T-lymphocyte function.⁵

Other factors besides immune dysfunction predispose to infection in persons with CKD. Thirty-nine percent of patients with end-stage kidney disease reported by the US Renal Data System are age 65 years or older,¹ and in other countries the percentage exceeds 50%. Besides advanced age, many CKD patients have underlying kidney or extrarenal autoimmune disease that requires therapeutic immunosuppression and other comorbidities that facilitate infection such as diabetes or heart failure. Dialysis patients and patients with cardiorenal syndrome

often have excess pulmonary fluid, which may impair alveolar bacterial clearance and contribute to the risk for pneumonia. Repeated breaks in the skin barrier by cannulation provide opportunity for bacterial invasion. Finally, dialysis patients frequently receive intravenous iron preparations, and iron overload is a risk factor for bacterial infection as a result of inhibition of monocyte and macrophage function.

Viral Infections

Viral hepatitis has been a scourge of dialysis ever since kidney replacement therapy became a routine treatment. Only rigorous infection control practices and active vaccination against hepatitis B virus (HBV) made large-scale HD safe. However, impaired immune defense was only one among other causes for the high prevalence of viral hepatitis in this patient group. The viruses can be easily transmitted nosocomially in HD patients without proper precautions. Nevertheless, the early outbreaks of HBV infection showed an abnormal clinical course related to the immune defect in CKD.⁶ The acute infection could sometimes only be detected by measurement of transaminases; the typical clinical syndrome of icterus, subfebrile temperatures, and malaise was often absent. Most affected dialysis patients developed chronic persistent infection, which occurs in only 10% of patients with normal kidney function.

The clinical manifestations of hepatitis C virus infection are not significantly influenced by the immune deficiency of kidney failure. The infection usually runs a subclinical course and becomes chronic even in the absence of CKD.

Vaccinations in Chronic Kidney Disease

Immune deficiency leads to a high rate of nonresponse to hepatitis B vaccination. Dialysis patients should always receive a double-dose vaccination and extended vaccination protocols ([Fig. 87.1](#)), but even with the most recent vaccines, some 20% of patients do not develop protective antibody levels.⁷ This impaired response led to the development of specifically adjuvanted vaccines or intracutaneous application protocols⁸ with improved efficiency in former nonresponders.

Other vaccinations are also affected by the immune defect in CKD. The Kidney Disease: Improving Global Outcomes (KDIGO) guideline⁹ recommends annual influenza vaccinations. Studies comparing vaccination efficiency in patients with CKD and healthy individuals have methodological limitations; therefore the extent to which immune dysfunction compromises these vaccination results is unclear. Most likely, the efficacy is lower in patients receiving dialysis. A recent systematic review on the protective effects with regard to hard endpoints such as mortality or hospitalization pointed out that the universal recommendation to vaccinate—albeit very plausible—has a weak evidence base.¹⁰ In contrast to the experience in the general population, enhanced-dose influenza vaccine does not seem to be superior to standard vaccines.¹¹

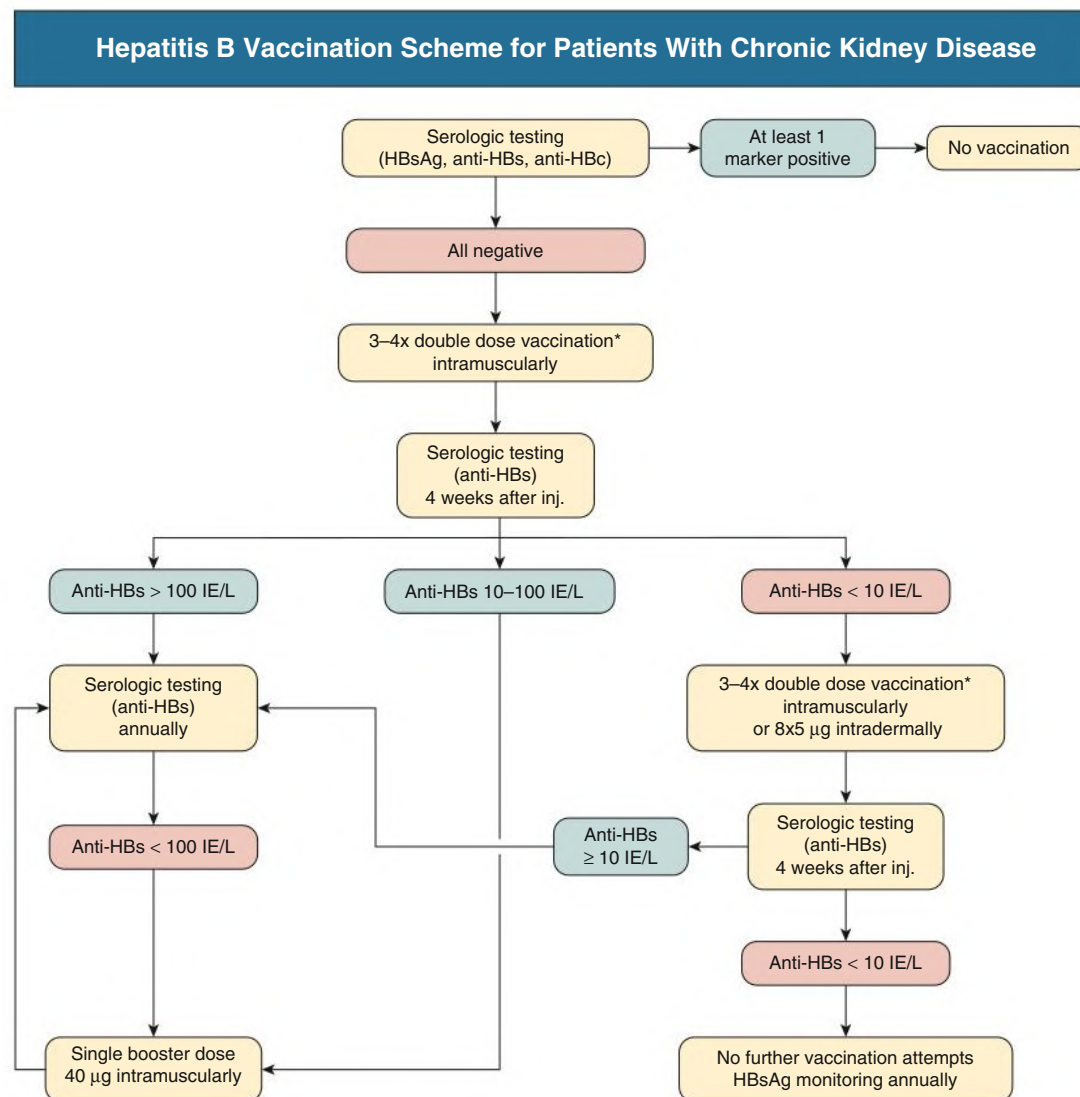


Fig. 87.1 Hepatitis B vaccination scheme for patients with chronic kidney disease (CKD). *The schedule varies with different vaccine preparations. HBsAg, Hepatitis B surface antigen.

Vaccines against herpes zoster have proven to be most effective in preventing zoster manifestations in the elderly. It seems to be justified to extend the general recommendation to vaccinate individuals older than 60 years to patients with CKD. The newer recombinant vaccine has not been studied in this patient group; however, the earlier attenuated vaccine led to excellent seroresponses in dialysis patients.¹²

Several studies have been published on SARS-CoV-2 vaccination in patients with CKD and on dialysis; this vaccination is highly desirable in these vulnerable populations (see [Chapter 59](#)).

In contrast to antiviral vaccinations, the efficacy of those against bacteria appear to be mildly reduced by CKD. Vaccination against pneumococcal infections is recommended⁹ for all adults with eGFR less than 30 mL/min/1.73 m². Evidence of efficacy in terms of antibody titers comes from very old studies or series with small patient numbers. Likely, maximum antibody titers are lower and protection lasts shorter in CKD patients,¹³ but data on hospitalization or mortality can be derived only from observational studies.¹⁴ Because multiple vaccine preparations are available, the congruence of local epidemiology and the vaccine coverage should be considered ([Fig. 87.2](#)). Newer conjugate vaccines have a rather narrow spectrum of serotype coverage while being highly effective in inducing seroprotection; thus there is an

argument to vaccinate CKD patients sequentially with conjugate vaccine first, followed in 6 to 12 months by the broad polysaccharide vaccine. Revaccination with the polysaccharide vaccine is recommended after 6 years.

Further recommendations for vaccination in CKD are given in [Table 87.1](#).

INFLAMMATION

Chronic inflammation is typical for patients with stage 4 or 5 CKD or on kidney replacement therapy, and C-reactive protein (CRP) and plasma cytokines increase steadily with every stage of CKD. The majority of HD and peritoneal dialysis patients periodically or permanently have CRP values above the normal limit despite the absence of clinical infection.¹⁵

Causes of Inflammation

Although early research in this field focused on dialysis membrane bio-incompatibility, contamination of dialysis fluids, or vascular access as causes of inflammation, chronic inflammation also occurs in patients with stages 3 and 4 CKD. One important reason may be the renal

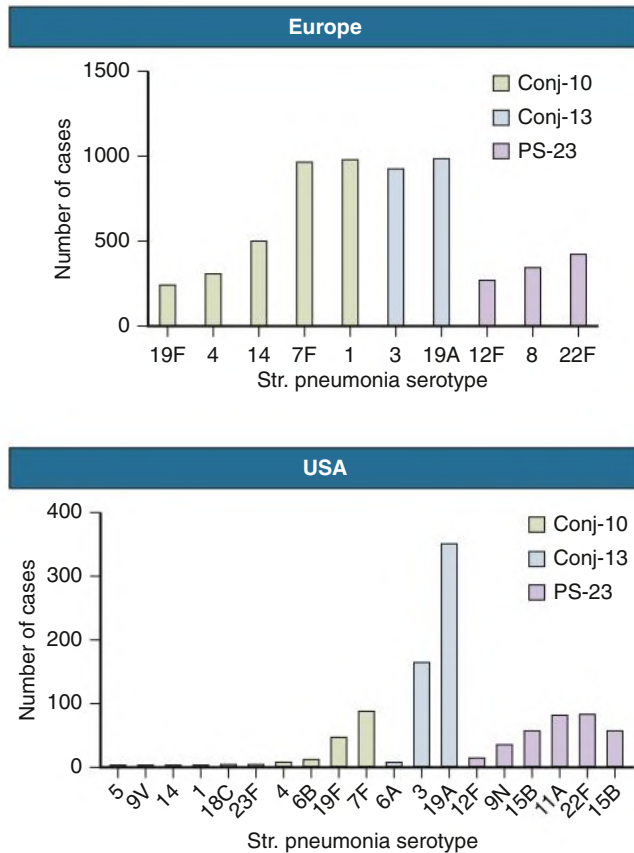


Fig. 87.2 *Streptococcus pneumoniae* serotypes in clinical isolates (as of 2010) from Europe (European Center of Disease Control) and United States (Centers for Disease Control and Prevention) in relation to vaccine serotype coverage. *Conj-10*, 10-valent conjugate vaccine; *Conj-13*, 13-valent conjugate vaccine; *PS-23*, 23-valent polysaccharide vaccine. Low isolate numbers for the serotypes covered by *Conj-10* in the United States are the result of successful vaccination programs.

TABLE 87.1 Vaccination Recommendations for Adult Patients With Chronic Kidney Disease

Vaccination Against	Recommendation
HBV	All patients susceptible to HBV infection, double dose (see Fig. 87.1)
Influenza	Annually according to WHO recommendation, standard dose
<i>Streptococcus pneumoniae</i>	All patients, conjugate vaccine, can be boosted by polysaccharide vaccine at 6 mo, revaccination after 6 yr
Tetanus	All patients according to recommendations for general population, standard dose
Diphtheria	All patients according to recommendations for general population, standard dose
Herpes zoster	All patients >60 years
Hepatitis A	Not routinely indicated, can be given prior to travel to affected areas

HBV, Hepatitis B virus; WHO, World Health Organization. Modified from Dinits-Pensy M, Forrest GN, Cross AS, Hise MK. The use of vaccines in adult patients with renal disease. *Am J Kidney Dis*. 2005;46:997–1011.

clearance of circulating proinflammatory cytokines.¹⁶ Impaired kidney function leads to prolonged cytokine serum half-life, thus enhancing and prolonging inflammatory episodes that usually would have abated quickly.

At least 30% of patients with advanced CKD have diabetes mellitus, a comorbidity that promotes inflammation both by itself and through diabetic complications such as ulcers. In addition, many patients with CKD have poor dental health, and periodontitis is associated with systemic inflammation.¹⁷

Whether inflammation is also related to classic uremic toxicity remains a matter of debate. Typical toxins with increased plasma levels are indoxyl sulfate, paracresyl sulfate, and trimethylamine N-oxide (TMAO). Across stages of CKD, plasma levels of TMAO show a positive correlation with inflammatory markers, and both decrease after kidney transplantation.¹⁸ Several studies show that serum levels of indoxyl sulfate or paracresyl sulfate increase with kidney dysfunction in parallel with inflammatory markers; however, a causal relationship remains to be proven.

The extent of inflammation is at least in part controlled by the genetic predisposition of the individual. This may be clinically relevant in patients with CKD because inherited single-nucleotide polymorphisms in the gene of the antiinflammatory cytokine interleukin (IL)-10 influence the quantity of its production in response to inflammatory stimuli and thus the way a patient can contain and limit inflammation.¹⁹

Consequences of Inflammation

Inflammation is closely associated with the occurrence of cardiovascular complications (see Chapter 85) and immune dysfunction. When using the response to HBV vaccination as a surrogate for clinical immune function, there is a close correlation between inflammation and decreased immunity. This is confirmed by the finding that the IL-10 gene polymorphism, which influences the extent of inflammation, also predicts vaccination responses in dialysis patients.¹⁹

IMMUNE CELL ABNORMALITIES

Monocytes

Monocytes are bone marrow–derived cells with specific functions in immune surveillance and antigen presentation. During their differentiation, they briefly circulate in the blood, produce cytokines, and then migrate into tissues to become macrophages.²⁰ Injury and infection lead to rapid enhancement of circulating monocyte numbers, which then migrate to the site of tissue injury and initiate the local immune response. Monocyte-derived macrophages are constituents of atherosclerotic plaques, and imaging studies have revealed that monocytes patrol along the vascular endothelium, where they detect endothelial defects.²¹

Monocyte subpopulations are defined by the expression of the lipopolysaccharide receptor CD14 and the immunoglobulin Fcγ receptor CD16.²² The classic monocytes only express CD14 and account for some 80% of all circulating monocytes in healthy individuals; intermediate monocytes express both CD14 and CD16 (~5%–7%), and nonclassic monocytes express high levels of CD16 while showing limited staining for CD14 (~10%–12%). Genetic analysis revealed that CD14⁺⁺CD16⁺ intermediate monocytes particularly express markers related to antigen presentation, inflammation, and angiogenesis.²³

In CKD, the intermediate and nonclassic monocyte subtypes are significantly expanded. This finding is related to the CV risk²⁴; patients with the highest rate of CD14⁺⁺CD16⁺ monocytes in blood had the lowest CV event-free survival.

Mouse models confirm differences between monocyte subpopulations in their ability to invade atherosclerotic plaques.²⁵ In CKD the expansion of proinflammatory monocyte subsets and their epidemiologic association with CV events make a causal role of these cells for atherosclerotic disease likely. Furthermore, monocytes also express components of the angiotensin system. The angiotensin-converting enzyme (ACE) that turns angiotensin I (Ang I) into vasoactive angiotensin II (Ang II) is expressed in atherosclerotic plaques and colocalizes with monocyte-derived macrophages.²⁶ ACE is also expressed on circulating monocytes, particularly on those with the CD14⁺⁺CD16⁺ phenotype.²⁷ Dialysis patients with high expression of ACE on intermediate monocytes have a dramatically enhanced CV mortality risk.²⁷

Expression of ACE on monocytes is strongly upregulated by the uremic milieu.²⁸ A consequence of this might be that monocytes transmigrate into the subendothelial space of arteries and provide high levels of ACE in the atherosclerotic plaque. The local production of Ang II through ACE is thought to contribute to further leukocyte attraction, inflammatory activation, and plaque growth. In addition, expression of ACE on monocytes alters their functional capacities. In vitro assays show that a higher expression of ACE leads to stronger endothelial adhesion and transmigration of the monocytes. This effect appears to be mediated via locally produced Ang II, because adhesion and transmigration could be inhibited by losartan.²⁸

Monocytes also express ACE-2, a peptidase that degrades Ang II to Ang1-7, a vasodilatory peptide. Whereas CKD leads to the overexpression of ACE, the ACE-2 enzyme is downregulated compared with healthy individuals.²⁹ Experimental overexpression of ACE-2 in a rodent model of atherosclerosis limited the progression of disease. These findings suggest that the uremic milieu alters monocyte function in a strongly proatherogenic way.

Monocytes are closely related to circulating blood dendritic cells. Dendritic cells are mainly found in organs and tissues, where they have strong capabilities in antigen presentation and activation of immune reactions. Their immature precursors circulate in blood in low numbers. There are different subtypes of circulating dendritic cells, but investigators have used different marker sets for their detection. This limits comparison among different studies, so that understanding of dendritic cell quantification and pathophysiology remains limited. There is a relation between elevated numbers of CD14⁺⁺CD16⁺ monocytes in the blood and the propensity of these cells to differentiate into dendritic cells in cell culture.³⁰ Other studies³¹ found significantly lower numbers of dendritic cells in advanced CKD compared with healthy controls. The finding may be related to cardiovascular disease, because studies in patients with coronary heart disease and normal kidney function also reported reduced circulating numbers of dendritic cells.

T Lymphocytes

Impaired vaccination responses against viral antigens such as HBV or influenza, as well as reduced skin reaction in the Mantoux test, result from impaired T-cell activation. T cells are an important component of the antigen-specific adaptive immune defense. Their activation depends on antigen-presenting cells (APCs) that present foreign antigens with major histocompatibility complex and provide important costimulatory signals. Only T cells with specificity of their T-cell receptor toward the particular antigen are activated and proliferate.

Major APCs are dendritic cells, and their precursors are monocytes. Early studies showed that proliferative T-cell responses are impaired in dialysis patients, and this impairment is directly correlated with nonresponses to HBV vaccination. Both replacement of the patient's APCs in in vitro assays with cells from healthy donors and overexpression of costimulatory molecules on the APCs normalize proliferation of T cells, indicating that the major defect leading to reduced

T-cell activation is in the APC.³² However, these rather crude assays did not consider T-cell subpopulations. Helper T cells express the surface marker CD4 and interact with various other cell types. The CD4⁺ helper T cell is particularly needed for activation of B cells for antigen-specific seroresponses to viral antigens, as in HBV vaccination. The CD8⁺ cytotoxic T cells are important for antiviral defense because they are able to lyse infected host cells. In CKD the relation of CD4⁺/CD8⁺ T cells is reduced.

CD4⁺ helper cells can be further distinguished into cells that mainly support cellular immune reactions (helper T cells Type 1, Th1) and others that are more important for immunoglobulin production by members of the B-cell lineage (Th2). These cell types differ in the pattern of cytokines they produce, with interferon- γ being the major cytokine of Th1 and IL-4 the main cytokine of Th2 cells. CKD leads to a marked deviation of T cell differentiation toward the Th1 phenotype.³³ Most likely the major cause is elevated production of IL-12 by monocytes and APCs in the context of their inflammatory activation.³³

Another T-lymphocyte subpopulation is regulatory T cells (Tregs) that are important for the downregulation of immune responses once the aim of an antiinfectious response is reached. They prevent ongoing inflammation and the development of autoimmunity. The typical Treg cells originate in the thymus and have a distinct pattern of surface molecule expression. In patients with CKD the number of circulating Treg cells is unaltered, but their capacity to downregulate CD4⁺ helper T-cell activity is impaired.³⁴

B Lymphocytes

Impaired vaccination efficacy in CKD suggests impaired function of immunoglobulin-producing B lymphocytes and plasma cells; thus a major dysfunction of this cell type might be expected. However, CKD patients have normal circulating immunoglobulin levels. Their B-cell lymphopenia is modest and probably not very clinically relevant. It is caused by a higher rate of apoptosis of these cells compared with healthy individuals.³⁵ Lymphopenia appears to result from reduced numbers of the majority of B-cell subpopulations (naive B cells, memory B cells, etc.).³⁶ Taken together, the alterations of B-lymphocytes in CKD appear to be less pronounced than alterations of other immune cell types. Impaired vaccination responses are caused by altered interaction of APCs, helper T lymphocytes, and the cytokine network in CKD rather than by abnormalities of B lymphocytes.

Granulocytes

Polymorphonuclear granulocytes (neutrophils) are components of the antigen-independent innate immune system. Their main activity is to kill and phagocytose invading pathogens via numerous enzymes that produce bactericidal substances. Among them are defensins, proteolytic enzymes, and enzymes that produce highly active oxygen species such as hypochlorous acid. In CKD patients, these nonspecific defense systems are highly activated, and the cells spontaneously release more reactive oxygen species.³⁷ Inflammation, activation of different cell types, antiinfectious defense, and vascular disease are closely interwoven. Thus, the elevation of oxygen species H₂O₂ and malondialdehyde has predictive value for CV events and mortality in CKD.³⁸

Another important function of granulocytes is phagocytosis, which is mildly compromised in CKD.³⁹ Further, chemotactic attraction of the cells to the location of infection is modified by substances such as leptin that are retained in CKD.⁴⁰

It is difficult to establish whether these abnormalities can be improved by dialysis. HD inevitably involves extensive contact between blood and foreign surfaces. When cellulose-based membranes were still in use, the activation of the complement system by these membranes led to marked depletion of circulating granulocytes within the

first 20 minutes of a dialysis session.⁴¹ Newer synthetic membranes lead to minimal complement activation, and the leukocyte drop is much less pronounced. These findings on immune cells relate to CKD specifically, and some primary diseases (especially diabetes mellitus) further influence immune cell function.

PLATELET DYSFUNCTION AND PLATELET INHIBITORS IN CHRONIC KIDNEY DISEASE

Normal hemostasis begins with platelet adhesion to vascular endothelium and requires a relatively vasoconstricted vessel wall, integrity of platelet glycoproteins (GPs), and a normal quantity of large molecular weight, multimeric von Willebrand factor (vWF) (Figs. 87.3 and 87.4). Main platelet GPs are GPIb, the platelet receptor for vWF, involved in platelet adhesion, and GPIIb/IIIa, the platelet receptor for fibrinogen, involved in platelet aggregation.

Under static conditions, GPIb and vWF have no affinity for each other. However, these molecules develop a specific affinity for each other at high shear stress, resulting in arterial platelet adhesion. Aggregated fibrinogen-platelet mesh acts as a trap for binding and activation of other plasma clotting factors. The exposure of the preceding clotting factors to tissue factors, present on damaged endothelial cells, catalyzes the conversion of prothrombin to thrombin, which converts fibrinogen to fibrin. Subsequent cross-linking of insoluble fibrin results in a stable hemostatic plug.

Hemorrhagic Diathesis and Uremic Platelet Dysfunction

Patients with CKD have a high risk of bleeding. This hemorrhagic diathesis frequently has cutaneous (easy bruising, ecchymoses, or prolonged hemorrhage from needle puncture or postoperative sites) and mucosal (epistaxis; gastrointestinal or gingival bleeding) manifestations. More dramatic—albeit infrequent—manifestations

are hemorrhagic pericarditis/hemopericardium, hemorrhagic pleural effusion/hemothorax, and intracranial and retroperitoneal bleeding.⁴²

This hemorrhagic diathesis is not reflected in a prolongation of the prothrombin time or the partial thromboplastin time. Similarly, even though platelet counts may be moderately decreased because of platelet consumption outperforming platelet production, severe thrombocytopenia is rarely seen in uremia; the occurrence of very low platelet counts therefore requires a thorough search for alternative causes.⁴³ Instead, platelet dysfunction is generally considered as the central contributor to the high bleeding risk, and a high number of pathophysiologic alterations have been suggested to contribute, comprising alterations in platelet function and structure, and extrinsic factors (see Table 87.1). Unfortunately, many studies on platelet functions in CKD date back to the early days of clinical nephrology, when clinical care and dialysis treatment were less sophisticated, and when less advanced laboratory methods were available for evaluation of hemostasis. For the various pathophysiologic alterations listed in Box 87.1, controversial data have been published.

In the search for uremic toxins inducing platelet dysfunction, a direct pathologic role of urea can be ruled out because no correlation exists between blood urea levels and bleeding time,⁴⁴ and individuals with high serum urea levels but otherwise normal kidney function have no bleeding tendency.⁴⁵

Among the different extrinsic factors that contribute to platelet dysfunction, the contribution of anemia has gained particular interest. Physiologically, erythrocytes occupy the center of a vessel, displacing platelets from the axial flow toward the vessel walls. This allows platelets to adhere to injured endothelial cells and initiate the formation of a platelet plug. In anemia, platelets are more dispersed, which impairs their adherence to the endothelium. Moreover, in CKD, red blood cells may affect coagulation by releasing adenosine diphosphate (ADP), by

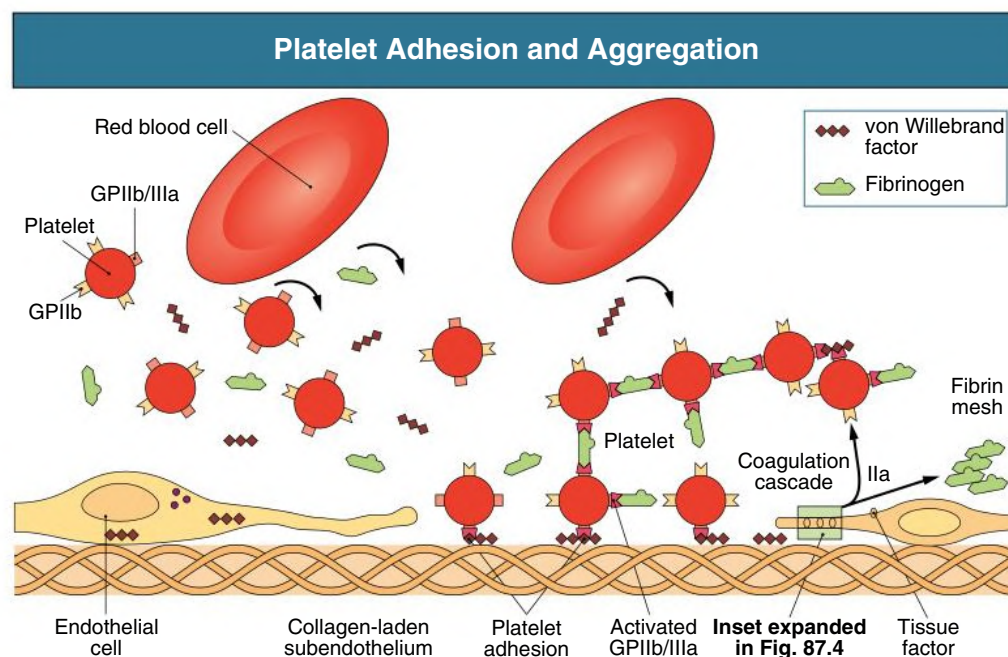


Fig. 87.3 Platelet Adhesion and Aggregation. Platelets are pushed peripherally toward the vascular wall by red blood cells traversing centrally through the bloodstream. Damage to the vessel wall results in a disruption of the nonthrombogenic endothelial cell lining and exposure of subendothelial structures. Whereas collagen supports initial platelet adhesion (and subsequent aggregation), von Willebrand factor (vWF) deposition on the subendothelium serves as the main anchor for platelet adhesion through platelet glycoprotein (GP) Ib receptor. Postadhesion conformational change in platelet GPIIb/IIIa receptor (fibrinogen or vWF receptor) results in interlinking platelet aggregation.

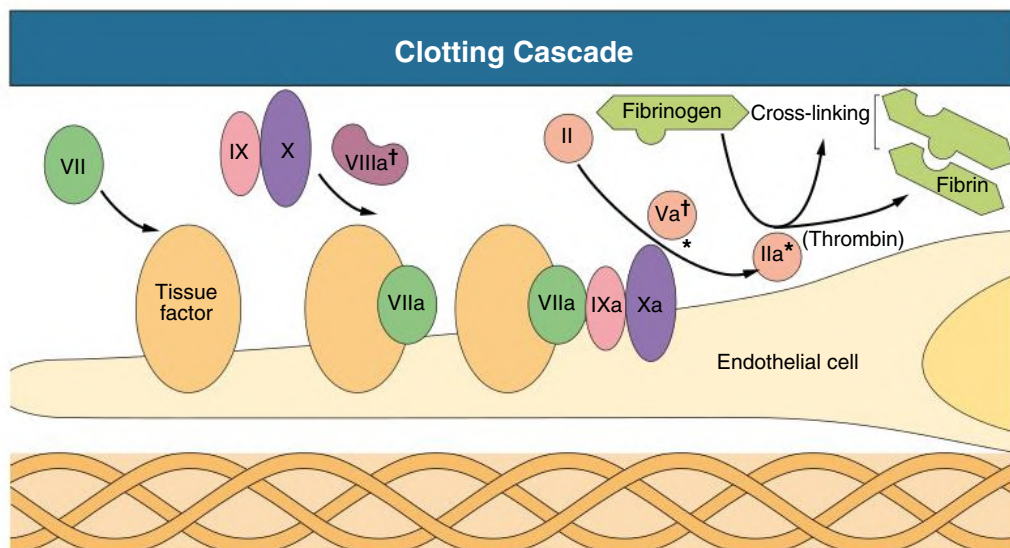


Fig. 87.4 Clotting Cascade. Expansion of the inset in Fig. 87.3 shows the clotting cascade that takes place at the damaged vessel wall. Exposure of subendothelial tissue factor, present on pericytes and fibroblasts, allows eventual activation of prothrombin (factor II) to thrombin. Thrombin converts fibrinogen to fibrin, activates fibrin cross-linking, stimulates further platelet aggregation, and activates anticoagulant protein C. Naturally occurring anticoagulants antithrombin III, protein C, and protein S help maintain control and counterbalance on coagulation. *Site of anticoagulant effect for antithrombin III. †Site of anticoagulant effect for protein C–protein S complex. (Courtesy James A. Sloand, MD, FACP, FASN, Baxter Healthcare Corporation, Deerfield, IL.)

BOX 87.1 Proposed Contributors to Platelet Dysfunction in Uremia

Intrinsic Factors That Contribute to Platelet Dysfunction

- Dysfunction of glycoprotein IIb/IIIa
- Abnormal expression of platelet glycoprotein
- Altered release of adenosine monophosphate and serotonin from platelet α -granules
- Faulty arachidonic acid and depressed prostaglandin metabolism, decreased platelet thromboxane A₂ generation
- Abnormal platelet cytoskeletal assembly with reduced incorporation of actin and diminished association of actin binding proteins (α -actinin and tropomyosin) with the cytoskeleton

Extrinsic Factors That Contribute to Platelet Dysfunction

- The action of uremic toxins
- Anemia
- Increased nitric oxide and cyclic guanosine monophosphate production
- Functional von Willebrand factor abnormalities
- Decreased platelet production
- Abnormal interactions between the platelet and the endothelium of the vessel wall

Modified from Berns JS. Platelet dysfunction in uremia. UpToDate. <https://www.uptodate.com/contents/platelet-dysfunction-in-uremia>.

inactivating PGI₂, and by scavenging nitric oxide (NO), which are all central regulators of platelet function.⁴³

Treatment of Uremic Platelet Dysfunction

Despite the previously discussed pathophysiologic considerations, few data establish the extent to which initiation of kidney replacement therapy reduces the risk for bleeding in CKD.

In the early decades of HD, the interaction of blood with cellulose-based dialyzer membranes resulted in complement activation and

transient thrombocytopenia during the dialysis procedure. When using more biocompatible dialyzer membranes, such complement-induced platelet reduction no longer has clinical relevance. Nonetheless, HD treatment may still affect bleeding disorders; although potentially removing uremic toxins, which affect platelet function, it requires systemic anticoagulation and exposes patients to the potential risk for heparin-induced thrombocytopenia (HIT). Moreover, HD may disrupt the platelet cytoskeleton, induce repeated platelet stress, decrease the percentage of RNA-rich platelets, and reduce the percentage of available reticulated platelets.⁴⁶ As RNA-rich and reticulated platelets are more able to be activated, the accumulation of less RNA-rich and less reticulated platelets indicates the presence of less reactive platelets. Although treatment of anemia may improve some parameters of platelet dysfunction,⁴⁶ it has not been demonstrated to ameliorate bleeding or risk for bleeding.

In summary, dialysis and anemia treatment will not completely normalize platelet function. Therefore, drug treatment should be considered for those patients who have active bleeding or are scheduled to undergo an invasive diagnostic or therapeutic procedure with bleeding risk.

Desmopressin

Desmopressin (1-deamino-8-D-arginine-vasopressin [DDAVP]) is a synthetic derivative of the antidiuretic hormone with little vasopressor activity. It acts by stimulating the release of large factor VIII–von Willebrand factor multimers from endothelial cells into the plasma and potentially by increasing the membrane glycoprotein expression of platelets.

At a dose of 0.3 μ g/kg (if given intravenously [in 50 mL of saline over 15–30 minutes] or subcutaneously), desmopressin may improve the bleeding time over the subsequent 4 to 8 hours. Typical side effects include water retention, hyponatremia, moderate thrombocytopenia, facial flushing, mild transient headache, nausea, abdominal cramps, and mild tachycardia; thrombotic events are rarely observed.⁴²

Although its efficacy on laboratory measures of platelet function is undisputed, its clinical efficacy for preventing or treating bleeding remains unproven. One study that recruited 162 participants without severely reduced kidney function who underwent ultrasound-guided kidney biopsy suggested that prebiopsy desmopressin administration may decrease the risk of bleeding and hematoma size.⁴⁷ However, given this rather limited evidence for a clinical benefit of desmopressin, there is broad consensus that desmopressin is not mandatory before a routine kidney biopsy. Additionally, depletion of endothelial stores of the factor VIII–von Willebrand factor multimers after a second DDAVP injection may result in tachyphylaxis, which precludes its chronic use.

Cryoprecipitate

In many uremic patients, cryoprecipitate may improve the bleeding time within 1 hour after infusion. This effect is supposed to be mediated by the provision of factor VIII–von Willebrand factor multimers, fibrinogen, and by other factors that enhance platelet aggregation. However, this effect is short lasting (4–24 hours). Moreover, cryoprecipitate may have infectious, hemorrhagic, and anaphylactoid complications, and not all patients respond to cryoprecipitate.^{42,43,46} Thus, use of cryoprecipitate should be limited to patients with life-threatening bleeding who are resistant to treatment with desmopressin and blood transfusions.

Tranexamic Acid

As an antifibrinolytic agent, tranexamic acid (TXA) is licensed for treatment of heavy bleeding, as well as before dental interventions in patients with coagulopathies. Its use in CKD has been reported in several (mostly small) cohort studies, which often focused on surrogate markers of hemorrhagic diathesis. Therefore, TXA may be considered for life-threatening bleeding events in CKD patients. However, it should be reserved for those in whom other treatments have failed to control bleeding, because kidney excretion is the main route of TXA clearance. TXA has an unpredictable pharmacokinetic profile in advanced CKD patients, who are at particular risk for neurologic side effects of TXA (i.e., seizures).⁴⁸

RECOMBINANT ACTIVATED FACTOR VII

Recombinant activated factor VII (rFVIIa) was developed for treatment of hemorrhage in individuals with hemophilia with antibodies inactivating factor VIII or IX. Because of the central role of activated factor VII in coagulation, rFVIIa has been used off-label in a variety of other severe bleeding disorders, including some case reports that claimed successful use of rFVIIa for treatment of bleeding in CKD patients.⁴³ This very limited evidence, together with a substantial risk for thromboembolic events, mandates very prudent use of rFVIIa, which should be considered only in very severe bleeding when other interventions have failed.

ESTROGENS

Estrogens may improve bleeding time in CKD patients in a dose-dependent manner. Their mode of action involves a reduced production of L-arginine, which is a precursor of NO. Thus, estrogens may lower elevated NO production in CKD patients and subsequently reduce NO-induced guanylyl cyclase stimulation and cyclic guanosine monophosphate (cGMP) synthesis, which will finally increase thromboxane A and ADP availability. Estrogens may additionally lower hemorrhagic diathesis by affecting production of coagulation factors and their inhibitors.

Estrogens may be given intravenously, orally, or cutaneously. Compared with other approaches, estrogen treatment may affect bleeding tendency for a prolonged time. However, its long-term safety has again not been assessed in prospective studies.

IMPLICATIONS FOR ANTIPLATELET AGENT THERAPY

Uremic platelet dysfunction and the increased bleeding risk also imply that treatment with antiplatelet agents—particularly aspirin, clopidogrel, prasugrel, and ticagrelor—may induce more bleeding events among CKD patients than in the general population. At the same time, CKD patients are at high CV risk, and use of aspirin or other antiplatelet agents may reduce their risk for myocardial infarctions and stroke.

A recent KDIGO guideline⁹ recommends that “adults with CKD at risk for atherosclerotic events be offered treatment with antiplatelet agents unless there is an increased bleeding risk that needs to be balanced against the possible CV benefits.” However, it remains unclear how to identify the patients who may benefit from antiplatelet agents, and risk scores from the general population work poorly among CKD patients.

Antiplatelet agents should not routinely be prescribed to all CKD patients because the increased risk for major bleeding appears to outweigh the CV benefits, at least in CKD patients with low risk for atherosclerotic events.⁴⁹ Their use among CKD patients with overt atherosclerotic vascular disease is strongly recommended (see [Chapter 85](#)). However, dual-antiplatelet therapy should be limited to the very early period after acute myocardial infarction or after coronary stenting among CKD patients, whose bleeding risk with dual-antiplatelet therapy is much higher than in patients with intact kidney function.

CIRCULATING COAGULATION FACTORS

Despite their elevated bleeding risk, patients with advanced CKD also may show features of a hypercoagulable state. Although general measures of the coagulation system—prothrombin time and partial thromboplastin time—are within normal ranges, venous thromboembolism (VTE) occurs more frequently in patients with low glomerular filtration rate and/or high albuminuria than among individuals with intact kidney function. A variety of factors may contribute, which include elevated levels of factor VIII and vWF, and a variety of comorbidities, including immobilization, congestive heart failure, and obesity. Iatrogenic factors may additionally play a part in CKD-associated hypercoagulation, such as drug treatment (erythropoietin, corticosteroids), intravascular interventions, and devices.

THERAPEUTIC INTERVENTION

Components of the coagulation system are targets of many drugs for prevention or treatment of thrombotic disease, including unfractionated and low-molecular-weight heparin (LMWH), vitamin K antagonists (VKAs), and non-vitamin K antagonist oral anticoagulants (NOACs).

First, heparin is routinely used during HD for preventing clotting in the extracorporeal circulation. Thus, nearly all HD patients are exposed to potential side effects of heparin, which include an increased bleeding risk and some less frequent adverse events, of which the development of HIT is the most serious complication. HIT is a clinical syndrome induced by antibodies that bind to heparin and platelet factor IV complexes on the platelet surface and thereby cause platelet activation. Clinically, HIT often manifests with arterial or venous

thrombosis; less frequent complications are venous limb gangrene, adrenal hemorrhagic necrosis, necrotizing skin lesions at heparin injection sites, and acute systemic reactions within a few minutes after exposure to unfractionated heparin or LMWH injections. Of note, many patients have heparin-dependent antibodies without clinically apparent HIT. Therefore, to avoid overdiagnosing HIT, immunologic tests should not be ordered in patients with a low clinical likelihood but focused on patients in whom HIT is suspected clinically (as suggested by the 4T score; Table 87.2). This is of particular importance for HD patients, in whom an incorrect diagnosis may result in the withdrawal of heparin treatment during HD and in the initiation of less established (and more expensive) anticoagulation strategies. In the absence of CKD-specific pathways, the diagnosis of HIT should follow recommendations from the general population. Here, the likelihood

of HIT should first be estimated with clinical prediction tools, and use of antigen assays for confirmation of HIT antibodies should focus on patients with high and intermediate clinical likelihood (Fig. 87.5). Importantly, these antigen assays for detection of HIT antibodies have a high sensitivity but poor specificity. Wherever available, functional assays (serotonin release assays or heparin-induced platelet activation) should be used to confirm positive findings from antigen assays in the majority of patients with suspected HIT.

Once a patient has a high or intermediate clinical probability of HIT, all heparin treatment must be stopped and alternative anticoagulation initiated. The most relevant treatment options are listed in Table 87.3. Of note, no VKA should be initiated at this stage because these agents may reduce activation of anticoagulatory protein C and thus further perpetuate the prothrombotic state.

TABLE 87.2 4Ts Score to Determine the Likelihood of Heparin-Induced Thrombocytopenia

	Score = 2	Score = 1	Score = 0
Thrombocytopenia: compare the highest platelet count within the sequence of declining platelet counts with the lowest count to determine the percent of platelet fall (select only one option)	>50% platelet fall <i>and</i> nadir of $\geq 20,000/\mu\text{L}$ <i>and</i> no surgery within preceding 3 days	>50% platelet fall <i>but</i> surgery within preceding 3 days <i>or</i> any combination of platelet fall and nadir that does not fit criteria for score 2 or score 0 (e.g., 30%–50% platelet fall or nadir 10–19)	<30% platelet fall Any platelet fall with nadir <10
Timing (of platelet count fall or thrombosis ^a) Day 0 = First day of most recent heparin exposure (select only one option)	Platelet fall day 5–10 after start of heparin Platelet fall within 1 day of start of heparin <i>and</i> exposure to heparin within past 5–30 days	Consistent with platelet fall days 5–10 but not clear (e.g., missing counts) Platelet fall within 1 day of start of heparin <i>and</i> exposure to heparin in past 31–100 days Platelet fall after day 10	Platelet fall day 4 or earlier without exposure to heparin in past 100 days
Thrombosis (or other clinical sequelae) (select only one option)	Confirmed new thrombosis (venous or arterial) Skin necrosis at injection site Anaphylactoid reaction to intravenous heparin bolus Adrenal hemorrhage	Recurrent venous thrombosis in a patient receiving therapeutic anticoagulants Suspected thrombosis (awaiting confirmation with imaging) Erythematous skin lesions at heparin injection sites	Thrombosis suspected
Other cause for thrombocytopenia^b (select only one option)	No alternative explanation for platelet fall is evident	Possible other cause is evident: sepsis without proven microbial source, thrombocytopenia associated with initiation of ventilator, other	<i>Probable other cause present:</i> Within 72 h of surgery Confirmed bacteremia/fungemia Chemotherapy or radiation within past 20 days DIC as a result of non-HIT cause PTP Platelet count <20 <i>and</i> given a drug implicated in causing D-ITP ^c (see list) Nonnecrotizing skin lesions at LMWH injection site (presumed DTH) Other

^aTiming of clinical sequelae, such as thrombocytopenia, thrombosis, or skin lesions.

^bTwo points if necrotizing heparin-induced skin lesions even if thrombocytopenia not present.

^cDrugs implicated in D-ITP: *Relatively common:* glycoprotein IIb/IIIa antagonists (abciximab, eptifibatide, tirofiban), quinine, quinidine, sulfa antibiotics, carbamazepine, vancomycin. *Less common:* actinomycin, amitriptyline, amoxicillin/piperacillin/nafticillin, cephalosporins (cefazolin, ceftazidime, ceftriaxone), celecoxib, ciprofloxacin, esomeprazole, fexofenadine, fentanyl, fusidic acid, furosemide, gold salts, levofloxacin, metronidazole, naproxen, oxaliplatin, phenytoin, propranolol, propoxyphene, ranitidine, rifampin, suramin, trimethoprim. *Note:* This is a partial list. See Fig. 87.5 for interpretation of score results.

DIC, Disseminated intravascular coagulation; D-ITP, drug-induced immune thrombocytopenia; DTH, delayed-type hypersensitivity; HIT, heparin-induced thrombocytopenia; LMWH, low-molecular-weight heparin; PTP, posttransfusion purpura.

Modified from *Antithrombotic therapy and prevention of thrombosis*, 9th ed: American College of Chest Physicians Evidence-Based Clinical Practice Guidelines. *Chest*. 2012;141(2 Suppl):e495S–e530S.

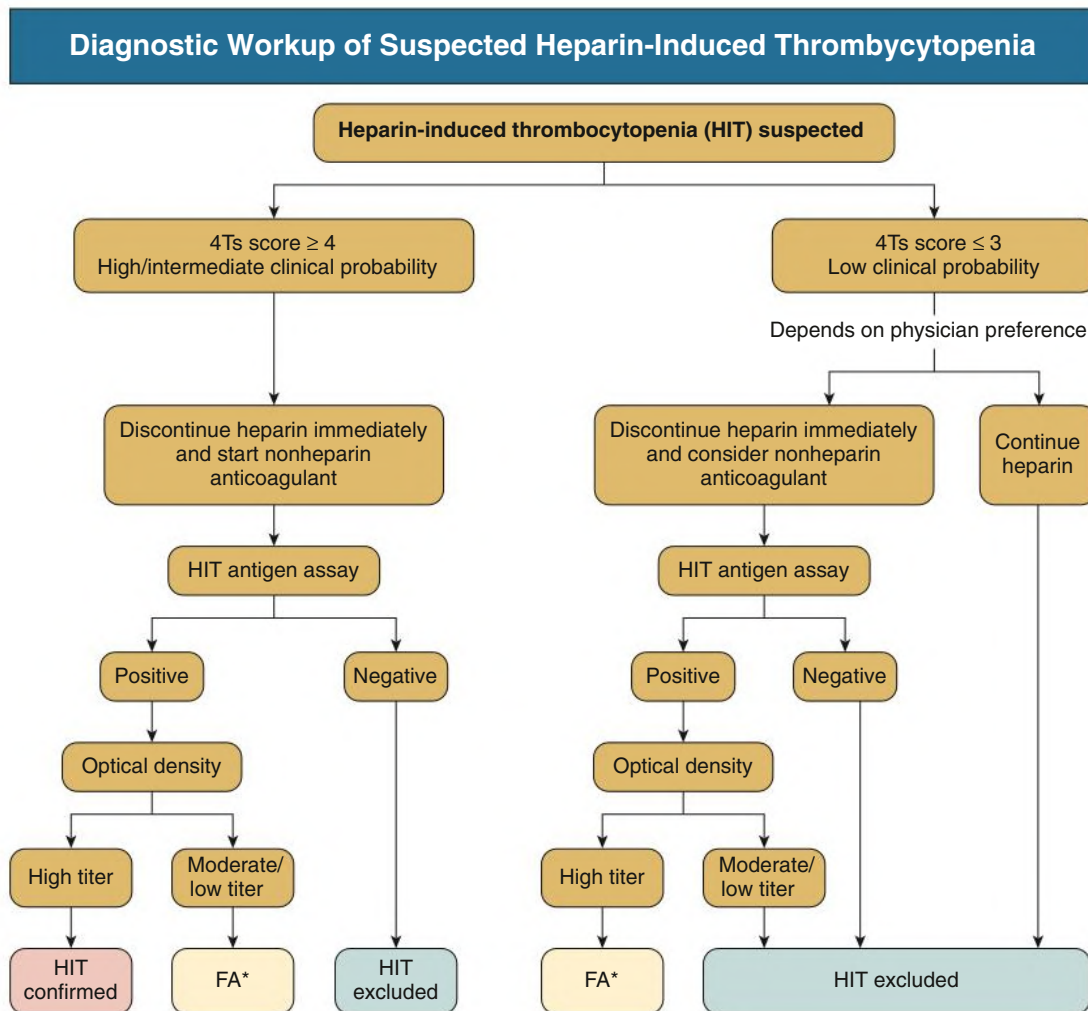


Fig. 87.5 Diagnostic Workup of Suspected Heparin-Induced Thrombocytopenia (HIT). Because of the high number of patients with antibodies against heparin without clinical HIT, a diagnostic algorithm is needed to have a high a priori likelihood of HIT before ordering antibody tests. There is no algorithm specific to patients with chronic kidney disease, and the diagnosis should be made as in patients with normal kidney function. *Confirm with functional assay (FA), if available. “4T” refers to a risk score based on four clinical parameters (thrombocytopenia, timing of platelet count fall or thrombosis, thrombosis [or other clinical sequelae], and other cause for thrombocytopenia). See Table 87.2 for a detailed description.

Additionally, anticoagulants are approved for prevention of thromboembolic stroke in patients with atrial fibrillation, for prevention and treatment of VTE (particularly deep vein thrombosis and pulmonary embolism), and for prevention of clotting in patients with mechanical heart valves. In the latter case, the need for lifelong anticoagulation with VKA is indisputable. Similarly, in patients with proximal deep vein thrombosis and symptomatic pulmonary embolism, anticoagulation is indicated for a minimum of 3 months. Recommendations on extended anticoagulation are mainly derived from the general population, in whom indefinite anticoagulation with either VKA or (preferably) NOACs (Table 87.4) may be considered in patients with unprovoked proximal deep vein thrombosis and symptomatic pulmonary embolism, whereas anticoagulation should be stopped after 3 months in patients with provoked VTE with one or more transient risk factors. In patients with active cancer and VTE, data from the general population suggest indefinite anticoagulation with LMWH or NOACs. When considering indefinite anticoagulation in patients with advanced CKD, their elevated bleeding risk must be considered. Therefore, if extended oral anticoagulation is decided after VTE in a CKD patient without cancer, a reduced dose of the NOACs apixaban (2.5 mg twice

daily) or rivaroxaban (10 mg once daily) should be considered after 6 months of therapeutic anticoagulation, which is supported both by evidence from randomized trials and by guidelines.⁵⁰ This is of particular importance, as dose reductions of NOACs in CKD patients with VTE do not fully follow recommendations for CKD patients with atrial fibrillation. Particularly, no dosage adjustment is recommended, but “use with caution” is recommended by the manufacturers of apixaban and rivaroxaban for any degree of reduced kidney function in the treatment of deep vein thrombosis and/or pulmonary embolism, whereas the manufacturer of edoxaban recommends 30 mg instead of (standard-dose) 60 mg edoxaban in patients with creatinine clearance of 15 to 50 mL/min.

For patients with advanced CKD, the use of anticoagulants in atrial fibrillation is controversial (see Chapter 85). In the general population, patients who have one or more risk factors for cerebral stroke or systemic embolization (defined as a CHA₂DS₂-VASc score of ≥1 in men and ≥2 in women) should receive oral anticoagulation. Similarly, use of warfarin (if adjusted to target an international normalized ratio [INR] of 2.0–3.0) has been shown to reduce stroke risk substantially in patients with stage 3a/3b CKD.⁵¹ In these patients, NOACs are at least

TABLE 87.3 Characteristics of Anticoagulants Used to Treat Patients With Heparin-Induced Thrombocytopenia

Characteristic	Argatroban	Danaparoid	Bivalirudin	Fondaparinux
Target	Thrombin	Factor Xa (predominantly)	Thrombin	Factor Xa
Elimination	Hepatobiliary	Renal	Enzymatic (80%)/renal (20%)	Renal
Approved for patients with HIT ^a	Treatment/PCI	Treatment	PCI/cardiac surgery	No
Method of administration	IV	IV, SC	IV	SC
Monitoring	aPTT ACT	Anti-Xa level	aPTT ACT or ECT (high doses)	Anti-Xa level
Effect on INR	+++	0	++	0
Immunologic features	None	5% cross-reactivity with HIT Ab ^b	Potentially cross-reactive with antilepirudin Ab	May cause HIT ^c
Antidote available	No	No	No	No
Dialyzable	20%	Yes	25%	20%

^aIn some countries (check with local health regulatory authorities).

^bClinical significance is uncertain and routine testing for cross-reactivity is not recommended.

^cCase reports only.

Ab, Antibodies; ACT, activated clotting time; aPTT, activated partial thromboplastin time; ECT, ecarin clotting time; HIT, heparin-induced thrombocytopenia; INR, international normalized ratio; IV, intravenous; PCI, percutaneous coronary intervention; SC, subcutaneous.

Modified from *Antithrombotic therapy and prevention of thrombosis*, 9th ed: American College of Chest Physicians Evidence-Based Clinical Practice Guidelines. *Chest*. 2012;141(2 Suppl):e495S–e530S.

TABLE 87.4 Absorption and Metabolism of Novel Oral Anticoagulants

	Dabigatran	Apixaban	Edoxaban	Rivaroxaban
Bioavailability	3%–7%	50%	62%	66% without food, almost 100% with food
Prodrug	Yes	No	No	No
Clearance nonrenal/renal, percent of absorbed dose ^a	20%/80%	73%/27%	50%/50%	65%/35%
Liver metabolism: CYP3A4 involved	No	Yes (elimination, moderate contribution ~25%)	Minimal (<4% of elimination)	Yes (hepatic elimination ~18%)
Absorption with food	No effect	No effect	6%–22% more; minimal effect on exposure	+39% more
Intake with food recommended?	No	No	No	Mandatory
Absorption with H2B/PPI	–12%–30% (not clinically relevant)	No effect	No effect	No effect
Dose if Asian ethnicity	+25%	No effect	No effect	No effect
GI tolerability	Dyspepsia 5%–10%	No problem	No problem	No problem
Elimination half-life	12–17 h	12 h	10–14 h	5–9 h (young), 11–13 h (elderly)

^aFor clarity, data are presented as single values, which are the midpoint of ranges as determined in different studies.

H2B, H₂ blocker; GI, gastrointestinal; PPI, proton pump inhibitor.

Modified from Steffel J, Verhamme P, Potpara TS, et al. 2018 European Heart Rhythm Association Practical Guide on the use of non-vitamin K antagonist oral anticoagulants in patients with atrial fibrillation. *Eur Heart J*. 2018;39(16):1330–1393.

as efficient as VKAs for prevention of thromboembolic events but may cause fewer intracerebral bleeding events. For patients with creatinine clearance less than 30 mL/min, few data demonstrate the efficacy (i.e., prevention of thromboembolic events) and safety (i.e., major bleeding events) of either VKAs or NOACs because most prospective clinical trials excluded patients with advanced CKD. The notable exception is very-low-dose edoxaban, which was tested in the ELDER-CARE study that also included patients with stage 4 CKD. In these patients, 15 mg edoxaban consistently protected against ischemic stroke without

causing a prohibitively high bleeding risk.⁵² Moreover, VKA accelerates vascular calcification^{53,54} and possibly also CKD progression, in particular in higher doses.⁵⁵ In dialysis patients with atrial fibrillation, retrospective cohort studies yielded conflicting findings on whether oral anticoagulation lowers stroke incidence (see [Chapter 85](#)).^{56,57}

For the time being, patients with atrial fibrillation and at least moderate risk for thromboembolism (defined by CHA₂DS₂-VASc score ≥2 for women, and ≥1 for men) should be offered NOACs if they have mild to moderate CKD (CKD 1–3b), unless they have prohibitive high

risk for bleeding. For stage 4 and 5 CKD patients with atrial fibrillation, lack of evidence precludes strong recommendations whether to use NOACs or no anticoagulation, with the potential alternative option of left atrial appendage closure. Based upon data from a first randomized clinical trial, VKAs appear inferior to dose-reduced NOACs in dialysis patients, but it is uncertain whether NOACs are preferable to no anticoagulation.⁵⁷

Both LMWH and NOACs require dose adjustment in patients with CKD because they undergo renal excretion. As of 2022, only one NOAC (apixaban) has been licensed for use among dialysis patients in the United States, but not yet in Europe. Compared with other NOACs, apixaban is characterized by the least accumulation in

CKD, but still requires (like all NOACs) dose adjustment for kidney function. Notably, pharmacokinetic studies (and subsequent recommendations on NOAC dosages) used creatinine clearance estimated with the Cockcroft-Gault equation. In clinical practice, eGFR is now mostly calculated using the Modification of Diet in Renal Disease (MDRD) or Chronic Kidney Disease–Epidemiology (CKD-EPI) formula. Substituting the latter for the former may cause large dosage errors, particularly when physicians fail to consider that the more recent equations yield estimates standardized for a body surface area of 1.73 m², and nonstandardized measures of kidney function (as provided by the Cockcroft-Gault equation) are suggested for dose adjustment.

SELF-ASSESSMENT QUESTIONS

- Which of the following diagnostic findings is reliable (sensitive and specific enough to be useful) in patients with CKD even in the presence of their typical immune dysfunction?
 - Fever as a sign of bacteremia
 - Radiologic infiltration on chest radiograph for pneumonia
 - Mendel-Mantoux skin reaction for tuberculosis
 - Blood culture for bacteremia
 - C-reactive protein for infection
- Viral hepatitis is a typical complication of hemodialysis treatment. Which statement correctly describes viral hepatitis in advanced CKD?
 - HBV infection leads to chronic infection in the majority of affected CKD patients.
 - Acute HBV infection in CKD patients is typically severe, characterized by jaundice and fever.
 - HBV infection can be easily prevented in CKD patients by vaccination, as recommended to the general population.
 - The clinical course of hepatitis C infection is largely different in CKD patients and those with normal kidney function.
 - Standard hygienic precautions have largely failed to reduce HBV transmission in dialysis centers. Only vaccination succeeded in preventing nosocomial transmission.
- The hemorrhagic diathesis in CKD patients is typically mirrored by which of the following?
 - Prolongation of the prothrombin time
 - Prolongation of the partial thromboplastin time
 - Severely thrombocytopenia
 - Hyperchromic anemia
 - None of these
- The spectrum of interventions that may allow reducing hemorrhages in advanced CKD does *not* include which of the following?
 - Desmopressin
 - Conjugated estrogens
 - Tranexamic acid
 - Cryoprecipitates
 - Clopidogrel

REFERENCES

- United States Renal Data System. 2016 USRDS Annual Data Report: *Epidemiology of Kidney Disease in the United States*. Bethesda, MD: National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases; 2016.
- James MT, Quan H, Tonelli M, et al. CKD and risk of hospitalization and death with pneumonia. *Am J Kidney Dis*. 2009;54:24–32.
- Ng JH, Hirsch JS, Wanchoo R, Sachdeva M, Sakhiya V, Hong S, et al. Outcomes of patients with end-stage kidney disease hospitalized with COVID-19. *Kidney Int*. 2020;98:1530–1539.
- Pisoni RL, Arrington CJ, Albert JM, et al. Facility hemodialysis vascular access use and mortality in countries participating in DOPPS: an instrumental variable analysis. *Am J Kidney Dis*. 2009;53:475–491.
- Sester M, Sester U, Clauser P, et al. Tuberculin skin testing underestimates a high prevalence of latent tuberculosis infection in hemodialysis patients. *Kidney Int*. 2004;65:1826–1834.
- London WT, DiFiglia M, Sutnick AI, Blumberg BS. An epidemic of hepatitis in a chronic hemodialysis unit. *N Engl J Med*. 1969;281:571–578.
- Janssen RS, Mangoo-Karim R, Pergola PE, et al. Immunogenicity and safety of an investigational hepatitis B vaccine with a toll-like receptor 9 agonist adjuvant (HBsAg-1018) compared with a licensed hepatitis B vaccine in patients with chronic kidney disease. *Vaccine*. 2013;31:5306–5313.
- Barracough KA, Wiggins KJ, Hawley CM, et al. Intradermal versus intramuscular hepatitis B vaccination in hemodialysis patients: a prospective open-label randomized controlled trial in nonresponders to primary vaccination. *Am J Kidney Dis*. 2009;54:95.
- Clinical KDIGO. Practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int Suppl*. 2013;3:1.
- Remschmidt C, Wichmann O, Harder T. Influenza vaccination in patients with end-stage renal disease: systematic review and assessment of quality of evidence related to vaccine efficacy, effectiveness, and safety. *BMC Med*. 2014;12:244.
- Butler AM, Layton JB, Dharnidharka VR, et al. Comparative effectiveness of high-dose versus standard-dose influenza vaccine among patients receiving maintenance hemodialysis. *Am J Kidney Dis*. 2020;75:72–83.
- Miller G, Schaefer H, Yoder S, et al. A randomized, placebo-controlled phase I trial of live, attenuated herpes zoster vaccine in subjects with end-stage renal disease immunized prior to renal transplantation. *Transpl Infect Dis*. 2018;20:e12874.
- Fuchshuber A, Kuhnemund O, Keuth B, et al. Pneumococcal vaccine in children and young adults with chronic renal disease. *Nephrol Dial Transplant*. 1996;11:468–473.
- Bond TC, Spaulding AC, Krisner J, McClellan W. Mortality of dialysis patients according to influenza and pneumococcal vaccination status. *Am J Kidney Dis*. 2012;60:959–965.
- Miyamoto T, Carrero JJ, Stenvinkel P. Inflammation as a risk factor and target for therapy in chronic kidney disease. *Curr Opin Nephrol Hypertens*. 2011;20:662–668.
- Kudo S, Goto H. Intrarenal handling of recombinant human interleukin-1 α in rats: mechanism for proximal tubular protein reabsorption. *J Interferon Cytokine Res*. 1999;19:1161–1168.
- Chen LP, Chiang CK, Chan CP, et al. Does periodontitis reflect inflammation and malnutrition status in hemodialysis patients? *Am J Kidney Dis*. 2006;47:815–822.
- Missailidis C, Hallqvist J, Qureshi AR, et al. Serum trimethylamine-N-oxide is strongly related to renal function and predicts Outcome in chronic kidney disease. *PLoS One*. 2016;11:e0141738.
- Girndt M, Sester U, Sester M, et al. The interleukin-10 promoter genotype determines clinical immune function in hemodialysis patients. *Kidney Int*. 2001;60:2385–2391.
- Ziegler-Heitbrock L. Blood monocytes and their subsets: established features and open Questions. *Front Immunol*. 2015;6:423.
- Auffray C, Fogg D, Garfa M, et al. Monitoring of blood vessels and tissues by a population of monocytes with patrolling behavior. *Science*. 2007;317:666–670.
- Ziegler-Heitbrock L, Ancuta P, Crowe S, et al. Nomenclature of monocytes and dendritic cells in blood. *Blood*. 2010;116:e74–e80.
- Zawada AM, Rogacev KS, Rotter B, et al. SuperSAGE evidence for CD14++CD16+ monocytes as a third monocyte subset. *Blood*. 2011;118:e50–e61.
- Heine GH, Ulrich C, Seibert E, et al. CD14(++)CD16(+) monocytes but not total monocyte numbers predict cardiovascular events in dialysis patients. *Kidney Int*. 2008;73:622–629.
- Tacke F, Alvarez D, Kaplan TJ, et al. Monocyte subsets differentially employ CCR2, CCR5, and CX3CR1 to accumulate within atherosclerotic plaques. *J Clin Invest*. 2007;117:185–194.
- Diet F, Pratt RE, Berry GJ, et al. Increased accumulation of tissue ACE in human atherosclerotic coronary artery disease. *Circulation*. 1996;94:2756–2767.
- Ulrich C, Heine GH, Seibert E, et al. Circulating monocyte subpopulations with high expression of angiotensin-converting enzyme predict mortality in patients with end-stage renal disease. *Nephrol Dial Transplant*. 2010;25:2265–2272.
- Trojanowicz B, Ulrich C, Seibert E, et al. Uremic conditions drive human monocytes to pro-atherogenic differentiation via an angiotensin-dependent mechanism. *PLoS One*. 2014;9:e102137.
- Trojanowicz B, Ulrich C, Kohler F, et al. Monocytic angiotensin-converting enzyme 2 relates to atherosclerosis in patients with chronic kidney disease. *Nephrol Dial Transplant*. 2016.
- Dopheide JF, Zeller GC, Kuhlmann M, et al. Differentiation of monocyte derived dendritic cells in end stage renal disease is skewed towards accelerated maturation. *Adv Clin Exp Med*. 2015;24:257–266.
- Agrawal S, Gollapudi P, Elahimehr R, et al. Effects of end-stage renal disease and haemodialysis on dendritic cell subsets and basal and LPS-stimulated cytokine production. *Nephrol Dial Transplant*. 2010;25:737–746.
- Girndt M, Köhler H, Schiedhelm-Weick E, et al. T-cell activation defect in hemodialysis patients: evidence for a role of the B7/CD28 pathway. *Kidney Int*. 1993;44:359–365.
- Sester U, Sester M, Hauk M, et al. T-cell activation follows Th1 rather than Th2 pattern in haemodialysis patients. *Nephrol Dial Transplant*. 2000;15:1217–1223.
- Afzali B, Edozie FC, Fazekasova H, et al. Comparison of regulatory T cells in hemodialysis patients and healthy controls: implications for cell therapy in transplantation. *Clin J Am Soc Nephrol*. 2013;8:1396–1405.
- Fernandez-Fresnedo G, Ramos MA, Gonzalez PM, et al. B lymphopenia in uremia is related to an accelerated in vitro apoptosis and dysregulation of Bcl-2. *Nephrol Dial Transplant*. 2000;15:502–510.
- Kim KW, Chung BH, Jeon EJ, et al. B cell-associated immune profiles in patients with end-stage renal disease (ESRD). *Exp Mol Med*. 2012;44:465–472.
- Yoon JW, Pahl MV, Vaziri ND. Spontaneous leukocyte activation and oxygen-free radical generation in end-stage renal disease. *Kidney Int*. 2007;71:167–172.
- Kalantar-Zadeh K, Brennan ML, Hazen SL. Serum myeloperoxidase and mortality in maintenance hemodialysis patients. *Am J Kidney Dis*. 2006;48:59–68.
- Anding K, Gross P, Rost JM, et al. The influence of uraemia and haemodialysis on neutrophil phagocytosis and antimicrobial killing. *Nephrol Dial Transplant*. 2003;18:2067–2073.
- Ottolento L, Gnerre P, Bertolotto M, et al. Leptin as a uremic toxin interferes with neutrophil chemotaxis. *J Am Soc Nephrol*. 2004;15:2366–2372.
- Craddock PR, Fehr J, Dalmaso AP, et al. Hemodialysis leukopenia. Pulmonary vascular leukostasis resulting from complement activation by dialyzer cellophane membranes. *J Clin Invest*. 1977;59:879–888.
- Kaw D, Malhotra D. Platelet dysfunction and end-stage renal disease. *Semin Dial*. 2006;19:317–322.
- Galbusera M, Remuzzi G, Boccardo P. Treatment of bleeding in dialysis patients. *Semin Dial*. 2009;22:279–286.
- Steiner RW, Coggins C, Carvalho AC. Bleeding time in uremia: a useful test to assess clinical bleeding. *Am J Hematol*. 1979;7:107–117.

45. Linthorst GE, Avis HJ, Levi M. Uremic thrombocytopenia is not about urea. *J Am Soc Nephrol*. 2010;21:753–755.
46. Hedges SJ, Dehoney SB, Hooper JS, et al. Evidence-based treatment recommendations for uremic bleeding. *Nat Clin Pract Nephrol*. 2007;3:138–153.
47. Manno C, Bonifati C, Torres DD, Campobasso N, Schena FP. Desmopressin acetate in percutaneous ultrasound-guided kidney biopsy: a randomized controlled trial. *Am J Kidney Dis*. 2011;57:850–855.
48. Ma TK, Chow KM, Kwan BC, et al. Manifestation of tranexamic acid toxicity in chronic kidney disease and kidney transplant patients: a report of four cases and review of literature. *Nephrology*. 2016.
49. Palmer SC, Di Micco L, Razavian M, et al. Antiplatelet therapy to prevent hemodialysis vascular access failure: systematic review and meta-analysis. *Am J Kidney Dis*. 2013;61:112–122.
50. Konstantinides SV, Meyer G, Becattini C, et al. ESC Scientific Document group. 2019 ESC Guidelines for the diagnosis and management of acute pulmonary embolism developed in collaboration with the European Respiratory Society (ERS). *Eur Heart J*. 2020;41:543–603.
51. Hart RG, Pearce LA, Asinger RW, Herzog CA. Warfarin in atrial fibrillation patients with moderate chronic kidney disease. *Clin J Am Soc Nephrol*. 2011;6:2599–2604.
52. Okumura K, Akao M, Yoshida T, et al. Low-dose edoxaban in very Elderly patients with atrial fibrillation. *N Engl J Med*. 2020;383:1735–1745.
53. van Gorp RH, Schurgers LJ. New Insights into the pros and cons of the clinical use of vitamin K antagonists (VKAs) versus direct oral anticoagulants (DOACs). *Nutrients*. 2015;7:9538–9557.
54. De Vriese AS, Caluwé R, Pyfferoen L, et al. Multicenter randomized controlled trial of vitamin K antagonist replacement by rivaroxaban with or without vitamin K₂ in hemodialysis patients with atrial fibrillation: the Valkyrie Study. *J Am Soc Nephrol*. 2020;31:186–19.
55. Bohm M, Ezekowitz MD, Connolly SJ, et al. Changes in renal function in patients with atrial fibrillation: an analysis from the RE-LY Trial. *J Am Coll Cardiol*. 2015;65:2481–2493.
56. Tan J, Liu S, Segal JB, et al. Warfarin use and stroke, bleeding and mortality risk in patients with end stage renal disease and atrial fibrillation: a systematic review and meta-analysis. *BMC Nephrol*. 2016;17:157.
57. Mavrakanas TA, Garlo K, Charytan DM. Apixaban versus no anticoagulation in patients undergoing long-term dialysis with incident atrial fibrillation. *Clin J Am Soc Nephrol*. 2020;15(8):1146–1154.